

Miller Indices

A crystal lattice may be regarded as the aggregate of different set of parallel planes passing through lattice points such as each set of parallel plane has particular orientation and direction with reference to co-ordinate axis.



- Miller Indices was first introduced in 1839 by British Scientist Miller.
- Miller Indices form a notation system in crystallography for planes in crystal (Bravais) lattices.
- In Particular, a family of lattice plane is determined by three integers h, k and l Miller Indices.

They are written as (hkl)

For ex:- (001)

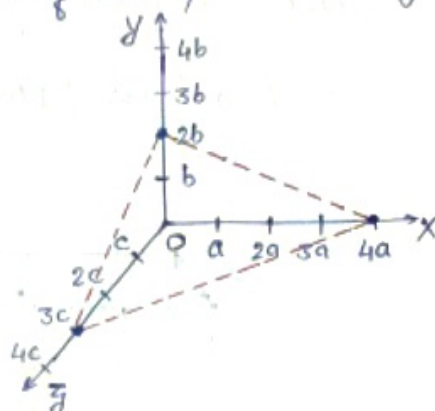
- The notation $\{hkl\}$ denotes the set of all planes that are equivalent to (hkl) by the symmetry of the lattice.

Note:- The Miller indices are defined with respect to any choice of unit cell and not only with respect to primitive basis vectors.

For ex:-

1. STEP:- Take a point as atom as origin and construct three co-ordinate axis. Find the intercepts of the plane along three co-ordinate axis.

$$\begin{array}{ccc} x & y & z \\ 4a & 2b & 3c \end{array}$$



- 2 STEP:- Express the intercepts as multiples of a, b, c .

$$\begin{array}{ccc} x & y & z \\ 4 & 2 & 3 \end{array}$$

- 3 STEP:- Take reciprocals of Intercepts.

$$\frac{1}{4} \quad \frac{1}{2} \quad \frac{1}{3}$$

- 4 STEP:- Reduce reciprocals of intercepts into smallest set of integers in the same ratio by multiplying with their L.C.M

$$\frac{1 \times 12}{4} \quad \frac{1 \times 12}{2} \quad \frac{1 \times 12}{3}$$

$$(3 \ 6 \ 4)$$

- 5 STEP:- Enclose the smallest set of integers in the paranthesis and we will get Miller Indices.

So Miller indices are (364)

Features

- In cubic crystal the direction L^r to (hkl) is represented by $[hkl]$ i.e $[hkl]$ is the line L^r to plane (hkl)
- No commas are introduced in b/w the indices of single digit i.e (364) but when miller indices of plane are of double digit, these are separated by commas (10, 5, 15).
- The Miller indices (266), (133) represents the same plane.